P7.1 (a) $\underline{d=1}$ $\quad \varepsilon_n = \frac{\hbar^2}{2m}\left(\frac{\pi n}{L}\right)^2 = \frac{\hbar^2 k^2}{2m}$

As usual, the wavenumber $k = \frac{\pi n}{L}$ is quantized.

Density of states appears when we transform summation over orbitals to integration over energy:

$$\sum_n X(n) = \int d\varepsilon\, D(\varepsilon)\, X(\varepsilon) \quad \text{for arbitrary } X.$$

But $\displaystyle\sum_{n=0}^{\infty} X(n) = 2\int_0^\infty dn\, X(n) = 2\int_0^\infty \frac{L}{\pi}\, dk\, X(k)$ → two spin states!

$$dk = d\sqrt{\frac{2m\varepsilon}{\hbar^2}} = \sqrt{\frac{2m}{\hbar^2}}\,\frac{d\varepsilon}{2\sqrt{\varepsilon}}$$

$$\Rightarrow \int_0^\infty \frac{2L}{\pi}\sqrt{\frac{2m}{\hbar^2}}\,\frac{d\varepsilon}{2\sqrt{\varepsilon}}\, X(\varepsilon) = \int d\varepsilon\, D_1(\varepsilon)\, X(\varepsilon)$$

$$D_1(\varepsilon) = \frac{L}{\pi}\sqrt{\frac{2m}{\hbar^2\varepsilon}}$$

(b) $\underline{d=2}$ $\quad \varepsilon_n = \frac{\hbar^2}{2m}\frac{\pi^2(n_x^2+n_y^2)}{L^2}$

$$\sum_n X(n) = \underset{\text{spin}}{(2)}\frac{L^2}{\pi^2}\underbrace{\int_0^\infty dk_x \int_0^\infty dk_y}_{\frac{1}{4}\,2\pi\int_0^\infty dk\cdot k}\; X(k_x,k_y) = \frac{L^2}{\pi}\int_0^\infty \sqrt{\frac{2m\varepsilon}{\hbar^2}}\sqrt{\frac{2m}{\hbar^2}}\,\frac{d\varepsilon}{2\sqrt{\varepsilon}}\, X =$$

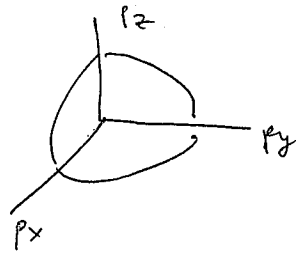
$$= \int \frac{L^2}{\pi}\,\frac{2m}{2\hbar^2}\, d\varepsilon\, X(\varepsilon) = \int d\varepsilon\, D_2(\varepsilon)\, X(\varepsilon)$$

$$D_2(\varepsilon) = \frac{L^2 m}{\pi\hbar^2}$$

P7.2 Relativistic Fermi gas : $\varepsilon = pc$

$$p = \frac{\pi\hbar}{L}\left(n_x^2+n_y^2+n_z^2\right)^{1/2} = \sqrt{p_x^2+p_y^2+p_z^2}$$

(a)



Number of electrons $N = \overset{\text{spin}}{2} \cdot \frac{1}{8} \cdot \frac{4\pi}{3} n_F^3 = \frac{\pi}{3} n_F^3$

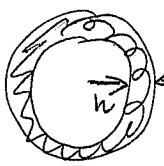
$$n_F = \left(\frac{3}{\pi} N\right)^{1/3}$$

Fermi momentum: $p_F = \frac{\pi\hbar}{L} n_F = \frac{\pi\hbar}{L}\left(\frac{3N}{\pi}\right)^{1/3} = \pi\hbar\left(\frac{3}{\pi}\frac{N}{L^3}\right)^{1/3}$

Fermi energy : $\varepsilon_F = c p_F$

$$\varepsilon_F = \pi\hbar c\left(\frac{3}{\pi} n\right)^{1/3} \quad \text{where } n = \frac{N}{L^3} = \frac{N}{V}.$$

(b) $U_0 = 2\sum_{n \le n_F} \varepsilon_n = 2 \cdot \frac{1}{8} \cdot 4\pi \int_0^{n_F} dn\, n^2 \left(\frac{\pi\hbar c}{L} n\right) =$



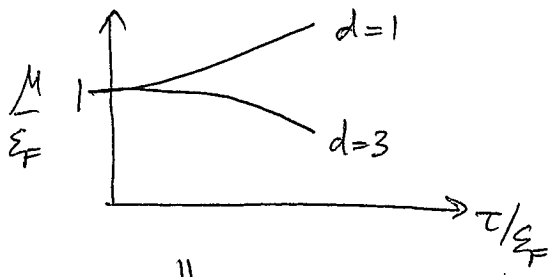
← volume of the shell $4\pi n^2 dn$

$$= \frac{\pi^2\hbar c}{L}\int_0^{n_F} dn\, n^3 = \frac{\pi^2\hbar c}{4L} n_F^4 = \frac{\pi^2\hbar c}{4L}\left(\frac{3}{\pi}N\right)^{4/3} =$$

$$= \frac{\pi^2\hbar c}{4L}\frac{3N}{\pi}\left(\frac{3N}{\pi}\right)^{1/3} = \frac{3\pi\hbar c N}{4L}\left(\frac{3N}{\pi}\right)^{1/3} = \frac{3\pi\hbar c N}{4}\left(\frac{3N}{\pi L^3}\right)^{1/3} =$$

$$= \frac{3\pi\hbar c N}{4}\frac{\varepsilon_F}{\pi\hbar c} = \frac{3}{4} N \varepsilon_F .$$

P7.4 Temperature dependence of μ in $d=1$ and $d=3$



Consult "Lecture Notes on Fermi gas at finite τ"

$$N_d = \int_0^\infty d\varepsilon\, D_d(\varepsilon)\, f(\varepsilon)$$ where $d=1,3$; $D_d(\varepsilon)$ = density of states (including spin)

$$D_1(\varepsilon) = \frac{L}{\pi}\left(\frac{2m}{\hbar^2\varepsilon}\right)^{1/2} \quad ; \quad D_3(\varepsilon) = \frac{L^3}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\sqrt{\varepsilon}$$

Introduce $G_d(\varepsilon)$ via: $\frac{d}{d\varepsilon}G_d(\varepsilon) = D_d(\varepsilon)$, then

$$N = \int_0^\infty d\varepsilon\, \frac{dG_d(\varepsilon)}{d\varepsilon}\, f(\varepsilon) = +\int_0^\infty d\varepsilon\, G_d(\varepsilon)\left(-\frac{df}{d\varepsilon}\right) =$$

$$= G_d(\mu) + \frac{1}{2}G_d''(\mu)\, I_2 \quad , \quad I_2 = \frac{\pi^2\tau^2}{3}$$

$\underline{d=3}$

$$\left.\begin{aligned} G_3(\varepsilon) &= \frac{L^3}{3\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\varepsilon^{3/2} \\ G''(\varepsilon) &= \frac{L^3}{4\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\frac{1}{\sqrt{\varepsilon}} \end{aligned}\right\} \text{ thus } \frac{\mu}{\varepsilon_F} = 1 - \frac{\pi^2\tau^2}{12\varepsilon_F^2}$$ as described in Lecture Notes.

$\underline{d=1}$ $\quad G_1(\varepsilon) = \frac{2L}{\pi}\left(\frac{2m}{\hbar^2}\right)^{1/2}\sqrt{\varepsilon}$, $G_1''(\varepsilon) = -\frac{L}{2\pi}\left(\frac{2m}{\hbar^2}\right)^{1/2}\frac{1}{\varepsilon^{3/2}}$

this sign is very important!

Thus $$N = \frac{2L}{\pi}\left(\frac{2m}{\hbar^2}\right)^{1/2}\sqrt{\mu} - \frac{L}{4\pi}\left(\frac{2m}{\hbar^2}\right)^{1/2}\frac{I_2}{\mu^{3/2}}$$

at $\tau=0$: $I_2 = 0 \Rightarrow \mu(\tau=0) = \varepsilon_F \Rightarrow N = \frac{2L}{\pi}\left(\frac{2m}{\hbar^2}\right)^{1/2}\sqrt{\varepsilon_F}$

$$\left(\frac{\pi N}{2L}\right)^2\frac{\hbar^2}{2m} = \varepsilon_F$$

Number of electrons in a piece of metal does not depend on τ (while chemical potential does), hence

$$N = \frac{2L}{\pi}\sqrt{\frac{2m}{\hbar^2}}\,\varepsilon_F^{1/2} = \frac{2L}{\pi}\sqrt{\frac{2m}{\hbar^2}}\,\mu^{1/2} - \frac{L}{4\pi}\sqrt{\frac{2m}{\hbar^2}}\,\frac{\pi^2\tau^2}{3\mu^{3/2}}$$

$$1 = \left(\frac{\mu}{\varepsilon_F}\right)^{1/2} - \frac{\pi^2\tau^2}{24}\cdot\frac{1}{\varepsilon_F^{1/2}\mu^{3/2}}$$

↳ can replace $\mu(\tau)$ by $\mu(\tau=0)=\varepsilon_F$ here.

$$\Rightarrow \left(\frac{\mu}{\varepsilon_F}\right)^{1/2} = 1 + \frac{\pi^2\tau^2}{24\varepsilon_F^2} \Rightarrow \boxed{\frac{\mu}{\varepsilon_F} = 1 + \frac{\pi^2\tau^2}{12\varepsilon_F^2} \quad \text{for } \underline{d=1}}$$

$$\left(\text{while } \frac{\mu}{\varepsilon_F} = 1 - \frac{\pi^2\tau^2}{12\varepsilon_F} \text{ for } \underline{d=3}\right).$$

$\boxed{P7.3}$ (a) From $U_0 = \frac{3\hbar^2}{10m}\left(\frac{\pi n_F}{L}\right)^2 N = \frac{3}{5}N\varepsilon_F$

$$P = -\frac{\partial U_0}{\partial V} = -\frac{\partial}{\partial V}\left(\frac{3\hbar^2}{10m}\frac{(\pi n_F)^2 N}{V^{2/3}}\right) = \frac{2}{3}\,\frac{3\hbar^2}{10m}\,\frac{(\pi n_F)^2 N}{V^{5/3}} =$$

$$= \left(\text{but } n_F = \left(\frac{3N}{\pi}\right)^{1/3}\right) = \frac{\hbar^2}{5m}\,\pi^2\left(\frac{3N}{\pi}\right)^{2/3}\frac{N}{V^{5/3}} = \frac{\hbar^2(3\pi^2)^{2/3}}{5m}\left(\frac{N}{V}\right)^{5/3}.$$

(b) From $C_V = \frac{1}{3}\pi^2\frac{3N}{2\varepsilon_F}\tau = \frac{\pi^2 N}{2\varepsilon_F}\tau$

and $C_V = \tau\left(\frac{\partial\sigma}{\partial\tau}\right)_V$ follows that $\left(\frac{\partial\sigma}{\partial\tau}\right)_V = \frac{\pi^2 N}{2\varepsilon_F}$

Thus $\sigma(\tau) = \frac{\pi^2 N}{2\varepsilon_F}\tau + \text{const}$

but $\sigma(\tau=0)=0 \Rightarrow \text{const}=0.$

$$\sigma(\tau) = \frac{\pi^2 N\tau}{2\varepsilon_F}.$$

P7.11 from P5.10 : $\langle (\Delta N)^2 \rangle = \tau \dfrac{\partial \langle N \rangle}{\partial \mu}$

$$\langle N \rangle = \frac{1}{e^{\frac{\varepsilon-\mu}{\tau}}+1}$$ for a single orbital ε.

Then $$\langle (\Delta N)^2 \rangle = \tau \frac{\partial}{\partial \mu} \frac{1}{e^{\frac{\varepsilon-\mu}{\tau}}+1} = -\tau \frac{-\frac{1}{\tau} e^{\frac{\varepsilon-\mu}{\tau}}}{\left(e^{\frac{\varepsilon-\mu}{\tau}}+1\right)^2} = \frac{e^{\frac{\varepsilon-\mu}{\tau}}+1-1}{\left(e^{\frac{\varepsilon-\mu}{\tau}}+1\right)^2} =$$

$$= \frac{1}{e^{\frac{\varepsilon-\mu}{\tau}}+1} - \frac{1}{\left(e^{\frac{\varepsilon-\mu}{\tau}}+1\right)^2} = \langle N \rangle - \langle N \rangle^2 = \langle N \rangle \left(1 - \langle N \rangle\right).$$

Note that fluctuations are absent when $\langle N \rangle = 0$ or 1, that is, for energy $\varepsilon \gg \varepsilon_F$ or $\varepsilon \ll \varepsilon_F$.